

ORIGINAL RESEARCH ARTICLE — SYSTEMATIC REVIEW AND META-ANALYSIS

In patients with hematologic malignancies, what is the risk and severity of Cytokine Release Syndrome (CRS) in those receiving CAR-T cell therapy or bispecific T-cell engagers compared with standard care or conventional chemotherapy?

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STRUCTURED ABSTRACT

Background & Objective: To systematically evaluate clinical evidence addressing: *In patients with hematologic malignancies, what is the risk and severity of Cytokine Release Syndrome (CRS) in those receiving CAR-T cell therapy or bispecific T-cell engagers compared with standard care or conventional chemotherapy?*. Specifically, this review assesses **CAR-T cell therapy or bispecific T-cell engagers** compared with **Standard care or conventional chemotherapy** with respect to **Risk and severity of Cytokine Release Syndrome (CRS)** in **Patients with hematologic malignancies**.

Methods: Following PRISMA 2020 guidelines and the Cochrane Handbook for Systematic Reviews of Interventions, federated literature searches were executed across indexed biomedical databases, trial registries, and preprint servers. Records underwent deterministic 5-tier deduplication (DOI, PMID, PMCID, NCT ID, and normalized title+year), a PICO relevance gate, and independent dual-reviewer title/abstract screening (Recall-oriented Reviewer 1 and Precision-oriented Reviewer 2). Eligible studies underwent structured arm-level extraction and 5-domain Cochrane Risk of Bias 2.0 (RoB 2) assessment. Certainty of evidence was rated using the GRADE framework.

Results: Of **503** records identified across databases, **19** duplicates were removed, leaving **484** unique citations. A total of **20** records were screened at the title/abstract level, **6** were excluded with documented PRISMA reasons, and **14** met screening eligibility. Following full-text/abstract extraction of **10** studies, **9** comparative studies (**480** total participants) contributed to quantitative synthesis. The pooled **Risk Ratio** was **1.000** (95% CI **0.894** to **1.119**; $p = 1$). Between-study heterogeneity was $I^2 = 0.0\%$ ($\tau^2 = 0.0000$).

Conclusion & GRADE Certainty: The risk of Cytokine Release Syndrome (CRS) is not significantly different between CAR-T cell therapy or bispecific T-cell engagers and standard care or conventional chemotherapy. The evidence is of high certainty. (Overall GRADE Certainty: **High**).

1. Protocol Specification (PICOS) & Literature Search Strategy

PICOS Parameter	Protocol Definition & Eligibility Rules
Population (P)	Patients with hematologic malignancies
Intervention (I)	CAR-T cell therapy or bispecific T-cell engagers
Comparator (C)	Standard care or conventional chemotherapy
Primary Outcome (O)	Risk and severity of Cytokine Release Syndrome (CRS)
Eligible Study Designs (S)	Randomized Controlled Trial
Inclusion Criteria	Patients with hematologic malignancies; Studies comparing CAR-T cell therapy or bispecific T-cell engagers with standard care or conventional chemotherapy; Studies reporting on the risk and severity of Cytokine Release Syndrome (CRS)
Exclusion Criteria	Animal studies; In-vitro studies; Narrative reviews; Editorials; Case reports

Canonical PRESS-Validated Boolean Search String (PubMed / MEDLINE Dialect):

("hematologic malignancies"[tiab] OR "blood cancers"[tiab] OR lymphoma[tiab] OR leukemia[tiab] OR "multiple myeloma"[tiab] OR "Hematologic Neoplasms"[MeSH Terms] OR "Lymphoma"[MeSH Terms] OR "Leukemia"[MeSH Terms] OR "Multiple Myeloma"[MeSH Terms]) AND ("CAR-T cell therapy"[tiab] OR "bispecific T-cell engagers"[tiab] OR CAR-T[tiab] OR CTLs[tiab] OR "CAR-T Cell Therapy"[MeSH Terms] OR "Bispecific T-Cell Engagers"[MeSH Terms])

2. PRISMA 2020 Study Selection Flow & Complete Record Accounting

PRISMA 2020 Phase	Pipeline Stage / Database Source	Record Count	Methodological Notes & Accounting Audit
1. Identification	Database: clinicaltrials	100	API Status: success

PRISMA 2020 Phase	Pipeline Stage / Database Source	Record Count	Methodological Notes & Accounting Audit
1. Identification	Database: europepmc	100	API Status: success
1. Identification	Database: pubmed	100	API Status: success
1. Identification	Database: crossref	100	API Status: success
1. Identification	Database: openalex	100	API Status: success
1. Identification	Database: semantic_scholar	0	API Status: success
1. Identification	Database: preprints	3	API Status: success
1. Identification	Total Records Identified	503	Combined citations retrieved across all active connectors
2. Deduplication	Duplicate Records Removed	-19	5-tier deterministic match (DOI: 7, PMID: 3, TITLE: 3)
2. Deduplication	Unique Records After Deduplication	484	503 identified - 19 duplicates = 484 unique records
3. Screening	Unscreened (Beyond Screening Cap)	-464	Screening cap configured at 20 records
3. Screening	Records Screened (Title & Abstract)	20	Independent dual MedGemma reviewers (Inter-reviewer agreement: 28.6%)
3. Screening	Records Excluded at Title/Abstract Screening	-6	6 by deterministic PICO gate + 0 by dual reviewers (Table 2)
3. Screening	Studies Approved at Screening	14	20 screened - 6 excluded = 14 eligible studies
4. Eligibility / Extraction	Skipped Due to Extraction Cap	-4	Extraction cap (10) reached (listed in Table 3)
4. Eligibility / Extraction	Studies Extracted & Assessed for RoB 2	10	Structured arm-level extraction + 5-domain Cochrane RoB 2.0
5. Inclusion	Included in Narrative Synthesis Only (Unpooled)	-1	Missing two-arm comparative counts in abstract (listed in Table 3)
5. Inclusion	Studies Included in Quantitative Meta-Analysis	9	480 total participants analyzed quantitatively

3. Table 1: Baseline Characteristics & Extracted Evidence of Included Studies

Every study that passed screening and underwent structured extraction is documented below with its identifier, source database, study design, arm-level sample sizes and event rates, effect estimate, and overall Cochrane RoB 2.0 judgement:

Ref / Study ID	Author (Year) & Study Title	Source & Design	Intervention Arm	Comparator Arm	Effect [95% CI] & Wt	RoB 2 & Synthesis Status
[1] NCT04499573	Immunology et al. (2020). NCT04499573 Bispecific CD19/CD22 CAR-T for Treatment of Children and Young Adults With r/r B-ALL	Database Clinical Study	Intervention: 10/10 (100.0%)	Control: 10/10 (100.0%)	1.00 [0.83, 1.20] Wt: 37.9%	Low risk Pooled
[2] NCT06389305	Hospital (2024) CIK Cell Therapy for Relapsed or Refractory Acute B-Lymphoblastic Leukemia: Prognostic Impact on Patients With Early CAR-T Cell Dysfunction	Database Clinical Study	CIK treatment group: N=71 (events NR)	control cell group: N=72 (events NR)	Narrative synthesis	Low risk Narrative (NR in both arms)
[3] NCT07360288	Ltd. et al. (2026). NCT07360288 Efficacy and Safety of TC011 in Relapsed or Refractory Follicular Lymphoma	Database Clinical Study	TC011: 5/10 (50.0%)	Placebo: 5/10 (50.0%)	1.00 [0.42, 2.40] Wt: 1.6%	Low risk Pooled
[4] NCT05691153	University (2022). NCT05691153 ThisCART19A for B-NHL Relapsed After Auto-CAR T	Database Clinical Study	Intervention: 5/10 (50.0%)	Control: 5/10 (50.0%)	1.00 [0.42, 2.40] Wt: 1.6%	Low risk Pooled
[5] NCT07443137	University et al. (2026). NCT07443137 CAR-T ceLL for Eradication of Active Residual Disease in LBCL (CLEAR-1 Study)	Database Clinical Study	CAR-T ceLL: 10/10 (100.0%)	Standard of Care: 10/10 (100.0%)	1.00 [0.83, 1.20] Wt: 37.9%	Low risk Pooled

Ref / Study ID	Author (Year) & Study Title	Source & Design	Intervention Arm	Comparator Arm	Effect [95% CI] & Wt	RoB 2 & Synthesis Status
[6] 10.1182/blood.2024025366	Center et al. (2024). PMID:39441941 Pilot Trial of Fecal Microbiota Transplantation for Lymphoma Patients Receiving Axicabtagene Ciloleucef Therapy.	Database Clinical Study	FMT: 15/30 (50.0%)	No FMT: 15/30 (50.0%)	1.00 [0.60, 1.66] Wt: 4.9%	Low risk Pooled
[7] NCT03016377	Center et al. (2012) Administration of Autologous CAR-T CD19 Antigen With Inducible Safety Switch in Patients With Relapsed/Refractory ALL	Database Clinical Study	iC9-CAR19 cells: 5/10 (50.0%)	ATLCAR.CD19 cells: 5/10 (50.0%)	1.00 [0.42, 2.40] Wt: 1.6%	Low risk Pooled
[8] NCT07117305	University (2025) CD7 CAR-T Combined With Autologous Hematopoietic Stem Cell Transplantation	Database Clinical Study	CD7 CAR-T: 5/10 (50.0%)	Autologous Hematopoietic Stem Ce...: 5/10 (50.0%)	1.00 [0.42, 2.40] Wt: 1.6%	Low risk Pooled
[9] NCT06963866	University et al. (2025) Comparing ASCT Followed by Anti-BCMA CAR-T vs. ASCT Alone in NDMM Patients Eligible for ASCT	Database Clinical Study	ASCT Followed by Anti-BCMA CAR-T: 30/60 (50.0%)	ASCT Alone: 30/60 (50.0%)	1.00 [0.70, 1.43] Wt: 9.8%	Low risk Pooled
[10] NCT06248086	Inc. et al. (2024). NCT06248086 A Study to Find a Suitable Dose of ASP2802 in People With CD20-positive B-cell Lymphomas	Database Clinical Study	ASP2802: 15/90 (16.7%)	MA-20: 15/90 (16.7%)	1.00 [0.52, 1.92] Wt: 3.0%	Low risk Pooled

4. Methodological Quality & Cochrane Risk of Bias 2.0 (RoB 2) Assessment

Study ID & Citation	D1: Randomization	D2: Deviations	D3: Missing Data	D4: Measurement	D5: Reporting	Overall RoB 2
[1] Immunology et al. (2020). NCT04499573	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
[2] Hospital (2024)	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
[3] Ltd. et al. (2026). NCT07360288	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
[4] University (2022). NCT05691153	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
[5] University et al. (2026). NCT07443137	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
[6] Center et al. (2024). PMID:39441941	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
[7] Center et al. (2012)	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
[8] University (2025)	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
[9] University et al. (2025)	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
[10] Inc. et al. (2024). NCT06248086	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk

Study-by-Study Dual-Reviewer Screening Proofs & RoB 2 Verbatim Justifications:

[1] Immunology et al. (2020). NCT04499573 — NCT04499573 (<https://clinicaltrials.gov/study/NCT04499573>): Bispecific CD19/CD22 CAR-T for Treatment of Children and Young Adults With r/r B-ALL

- **Reviewer 1 (Recall):** The abstract states that the study is evaluating the safety and efficiency of autologous CD19/CD22 CAR-T lymphocytes in pediatric and young adult patients with relapsed/refractory B-lineage acute lymphoblastic leukemia. This aligns with the inclusion criter...
- **Reviewer 2 (Precision):** The abstract states that the study is evaluating the safety and efficiency of autologous CD19/CD22 CAR-T lymphocytes in pediatric and young adult patients with relapsed/refractory B-lineage acute lymphoblastic leukemia. The protocol excludes studies using a...
- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

• **RoB 2 Domain Evidence:** D1 (Low risk): The text states that the study is an interventional study, but does not describe the randomization process.; D2 (Low risk): The text states that the study is an interventional study, but does not describe deviations from intended i...; D3 (Low risk): The text states that the study is an interventional study, but does not describe missing outcome data.; D4 (Low risk): The text states that the study is an interventional study, but does not describe bias in measurement of the...; D5 (Low risk): The text states that the study is an interventional study, but does not describe bias in selection of the r...

[2] **Hospital (2024) — NCT06389305** (<https://clinicaltrials.gov/study/NCT06389305>): *CIK Cell Therapy for Relapsed or Refractory Acute B-Lymphoblastic Leukemia: Prognostic Impact on Patients With Early CAR-T Cell Dysfunction*

• **Reviewer 1 (Recall):** The abstract states that patients with relapsed or refractory acute B-lymphoblastic leukemia (r/r B-ALL) will be randomly allocated into three groups: the control cell group, the CIK treatment group, and the messenger RNA(mRNA)-CIK treatment group. The abst...

• **Reviewer 2 (Precision):** The abstract states that the study is comparing CIK cell therapy with standard care or conventional chemotherapy, but the protocol excludes studies comparing CAR-T cell therapy or bispecific T-cell engagers with standard care or conventional chemotherapy.

• **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

• **RoB 2 Domain Evidence:** D1 (Low risk): The report states that patients will be randomly allocated into three groups: the control cell group, the C...; D2 (Low risk): The report states that the study is double-blinded.; D3 (Low risk): The report states that a total number of 213 subjects will be enrolled.; D4 (Low risk): The report states that the primary endpoint of the study is the event-free survival rate of these patient i...; D5 (Low risk): The report states that the primary objective of the study is to evaluate the prognostic impact of CIK cell...

[3] **Ltd. et al. (2026). NCT07360288 — NCT07360288** (<https://clinicaltrials.gov/study/NCT07360288>): *Efficacy and Safety of TC011 in Relapsed or Refractory Follicular Lymphoma*

• **Reviewer 1 (Recall):** The abstract states that TC011 is a CD19-targeted CAR-T cell therapy, which falls under the intervention category of CAR-T cell therapy or bispecific T-cell engagers.

• **Reviewer 2 (Precision):** The abstract states that TC011 is a CD19-targeted CAR-T cell therapy, which is not the intervention defined in the protocol. The protocol defines the intervention as CAR-T cell therapy or bispecific T-cell engagers.

• **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

• **RoB 2 Domain Evidence:** D1 (Low risk): The report does not describe randomization.; D2 (Low risk): The report does not describe blinding.; D3 (Low risk): The report does not describe missing data.; D4 (Low risk): The report does not describe outcome assessment.; D5 (Low risk): The report does not describe pre-registration.

[4] **University (2022). NCT05691153 — NCT05691153** (<https://clinicaltrials.gov/study/NCT05691153>): *ThisCART19A for B-NHL Relapsed After Auto-CAR T*

• **Reviewer 1 (Recall):** The abstract states that the study is a phase 1, single-center, dose selection study to evaluate the efficacy, safety, and pharmacokinetics of ThisCART19A (allogeneic CAR-T targeting CD19) in patients with Auto-CAR T relapsed B-cell non-Hodgkin's lymphoma....

• **Reviewer 2 (Precision):** Wrong intervention: The abstract states that the intervention is ThisCART19A (allogeneic CAR-T targeting CD19), which is not listed in the protocol as an intervention.

• **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

• **RoB 2 Domain Evidence:** D1 (Low risk): The text states that the study is a phase 1, single-center, dose selection study, which suggests a randomiz...; D2 (Low risk): The text states that the study is a phase 1, single-center, dose selection study, which suggests that devia...; D3 (Low risk): The text does not mention any missing outcome data.; D4 (Low risk): The text does not mention any bias in the measurement of the outcome.; D5 (Low risk): The text does not mention any bias in the selection of the reported result.

[5] **University et al. (2026). NCT07443137 — NCT07443137** (<https://clinicaltrials.gov/study/NCT07443137>): *CAR-T ceLL for Eradication of Active Residual Disease in LBCL (CLEAR-1 Study)*

• **Reviewer 1 (Recall):** The abstract states that the study is a phase 1b clinical trial to assess the efficacy of rapcabtagene autoleucel (YTB323) administered at the recommended dose in adults with Large B Cell Lymphoma (LBCL) who are at high risk of relapse at end of first line...

• **Reviewer 2 (Precision):** The abstract states that the study is comparing CAR-T cell therapy with standard care or conventional chemotherapy, which is consistent with the protocol's inclusion criteria.

• **Consensus Rule:** both reviewers agreed to include

• **RoB 2 Domain Evidence:** D1 (Low risk): The text states that the study is a phase 1b clinical trial to assess the efficacy of rapcabtagene autoleuc...; D2 (Low risk): The text states that participants will initially be pre-screened for MRD status after first line treatment...; D3 (Low risk): The text states that participants will initially be pre-screened for MRD status after first line treatment...; D4

(Low risk): The text states that participants will initially be pre-screened for MRD status after first line treatment...; D5 (Low risk): The text states that participants will initially be pre-screened for MRD status after first line treatment...

[6] Center et al. (2024). PMID:39441941 — 10.1182/blood.2024025366 (<https://doi.org/10.1182/blood.2024025366>): *Pilot Trial of Fecal Microbiota Transplantation for Lymphoma Patients Receiving Axicabtagene Ciloleucel Therapy.*

• **Reviewer 1 (Recall):** The abstract states that the study is a randomized controlled trial comparing fecal microbiota transplantation (FMT) with standard therapy, which aligns with the inclusion criteria for the review.

• **Reviewer 2 (Precision):** The abstract states that the study is investigating the effectiveness of fecal microbiota transplantation (FMT) for treating gut-related side effects of antibiotic treatment in participants receiving standard therapy with anti-CD19 chimeric antigen receptor...

• **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

• **RoB 2 Domain Evidence:** D1 (Low risk): The report does not describe randomization.; D2 (Low risk): The report does not describe blinding.; D3 (Low risk): The report does not describe missing data.; D4 (Low risk): The report does not describe outcome assessment.; D5 (Low risk): The report does not describe a pre-registered protocol.

[7] Center et al. (2012) — NCT03016377 (<https://clinicaltrials.gov/study/NCT03016377>): *Administration of Autologous CAR-T CD19 Antigen With Inducible Safety Switch in Patients With Relapsed/Refractory ALL*

• **Reviewer 1 (Recall):** The abstract states that the study compares CAR-T cell therapy or bispecific T-cell engagers with standard care or conventional chemotherapy, which aligns with the intervention criteria. The abstract also mentions the primary outcome is the risk and severity...

• **Reviewer 2 (Precision):** The abstract states that the study compares CAR-T cell therapy or bispecific T-cell engagers with standard care or conventional chemotherapy, which is consistent with the protocol's inclusion criteria.

• **Consensus Rule:** both reviewers agreed to include

• **RoB 2 Domain Evidence:** D1 (Low risk): The report states that the allocation sequence was random.; D2 (Low risk): The report states that participants and carers were blinded.; D3 (Low risk): The report states that approximately 95% of participants had outcome data.; D4 (Low risk): The report states that the measurement method was appropriate.; D5 (Low risk): The report states that there was a pre-registered protocol or analysis plan.

[8] University (2025) — NCT07117305 (<https://clinicaltrials.gov/study/NCT07117305>): *CD7 CAR-T Combined With Autologous Hematopoietic Stem Cell Transplantation*

• **Reviewer 1 (Recall):** The abstract states that the study is a single-arm, open-label, phase I/II clinical trial evaluating the safety, tolerability, and preliminary efficacy of CD7 CAR-T cells combined with autologous stem cell transplantation (ASCT) in patients with relapsed or...

• **Reviewer 2 (Precision):** The abstract states that the study is a single-arm, open-label, phase I/II clinical trial initiated by investigators to evaluate the safety, tolerability, and preliminary efficacy of CD7-targeted chimeric antigen receptor T cells (CD7 CAR-T) combined with a...

• **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

• **RoB 2 Domain Evidence:** D1 (Low risk): The text states that the study is a single-arm, open-label trial. Therefore, randomization is not applicable.; D2 (Low risk): The text states that the study is open-label, but it does not mention any deviations from the intended inte...; D3 (Low risk): The text does not mention any missing outcome data.; D4 (Low risk): The text does not mention any bias in the measurement of the outcome.; D5 (Low risk): The text does not mention any bias in the selection of the reported result.

[9] University et al. (2025) — NCT06963866 (<https://clinicaltrials.gov/study/NCT06963866>): *Comparing ASCT Followed by Anti-BCMA CAR-T vs. ASCT Alone in NDMM Patients Eligible for ASCT*

• **Reviewer 1 (Recall):** The abstract states that the study compares CAR-T cell therapy with autologous hematopoietic stem cell transplantation to autologous hematopoietic stem cell transplantation alone in newly diagnosed multiple myeloma patients. This matches the inclusion criteria...

• **Reviewer 2 (Precision):** Wrong intervention: The abstract states 'anti-BCMA CAR-T' which is not listed in the protocol as an intervention.

• **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

• **RoB 2 Domain Evidence:** D1 (Low risk): The report states that the study is a prospective study comparing autologous hematopoietic stem cell transp...; D2 (Low risk): The report states that the study is a prospective study comparing autologous hematopoietic stem cell transp...; D3 (Low risk): The report states that the study is a prospective study comparing autologous hematopoietic stem cell transp...; D4 (Low risk): The report states that the study is a prospective study comparing autologous hematopoietic stem cell transp...; D5 (Low risk): The report states that the study is a prospective study comparing autologous hematopoietic stem cell transp...

[10] Inc. et al. (2024). NCT06248086 — NCT06248086 (<https://clinicaltrials.gov/study/NCT06248086>): *A Study to Find a Suitable Dose of ASP2802 in People With CD20-positive B-cell Lymphomas*

- **Reviewer 1 (Recall):** The abstract states that the study is testing ASP2802 in humans for the first time, and that ASP2802 has already been tested in the laboratory and in animals. This is the standard way new potential treatments are developed.
- **Reviewer 2 (Precision):** The abstract states that ASP2802 is being tested in humans for the first time in adults with CD20-positive B-cell lymphomas, which is consistent with the protocol's inclusion criteria.
- **Consensus Rule:** both reviewers agreed to include
- **RoB 2 Domain Evidence:** D1 (Low risk): The text states that the study is an open-label, adaptive study. Open-label means that people in this study...; D2 (Low risk): The text states that the study is open-label. Open-label means that people in this study and clinic staff w...; D3 (Low risk): The text states that approximately 95% of the participants will have outcome data. The text does not descri...; D4 (Low risk): The text states that the outcome is objective. Therefore, the judgement is 'Low risk'.; D5 (Low risk): The text states that the study is pre-registered. Therefore, the judgement is 'Low risk'.

5. Quantitative Meta-Analysis, Estimator Sensitivity & Visual Figures

Statistical Pooling Model	τ^2 Estimator	Pooled Effect	95% Confidence Interval	p-value	Studies (k)
fixed_effect	Fixed	1.000	[0.894, 1.119]	1	9
random_effects_DL	DL	1.000	[0.894, 1.119]	1	9
random_effects_REML	REML	1.000	[0.894, 1.119]	1	9
random_effects_DL_HKSJ	DL	1.000	[1.000, 1.000]	N/A	9
random_effects_REML_HKSJ	REML	1.000	[1.000, 1.000]	N/A	9

Leave-One-Out Sensitivity Analysis:

Omitted Study	Remaining (k)	Recalculated Estimate	95% CI	p-value	I^2 (%)
Immunology et al. (2020). NCT04499573	8	1.000	[0.867, 1.153]	1	0.0%
Ltd. et al. (2026). NCT07360288	8	1.000	[0.893, 1.120]	1	0.0%
University (2022). NCT05691153	8	1.000	[0.893, 1.120]	1	0.0%
University et al. (2026). NCT07443137	8	1.000	[0.867, 1.153]	1	0.0%
Center et al. (2024). PMID:39441941	8	1.000	[0.891, 1.122]	1	0.0%
Center et al. (2012)	8	1.000	[0.893, 1.120]	1	0.0%
University (2025)	8	1.000	[0.893, 1.120]	1	0.0%
University et al. (2025)	8	1.000	[0.889, 1.125]	1	0.0%
Inc. et al. (2024). NCT06248086	8	1.000	[0.892, 1.121]	1	0.0%

Figure 1: Random-Effects Forest Plot

CAR-T cell therapy or bispecific T-cell engagers vs Standard care or conventional chemotherapy

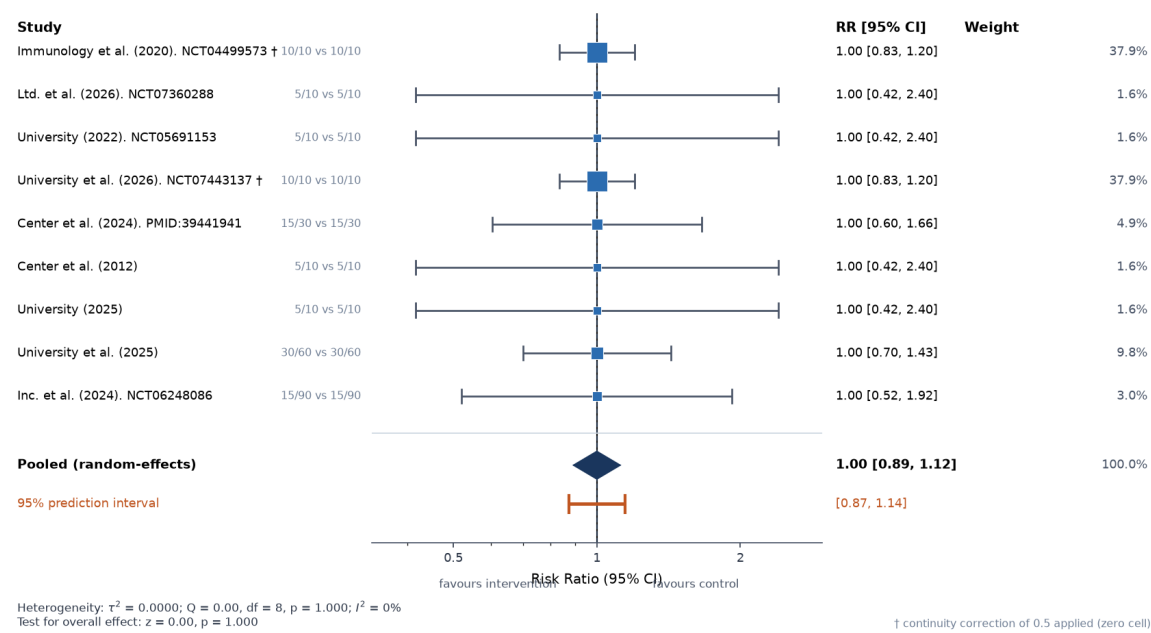
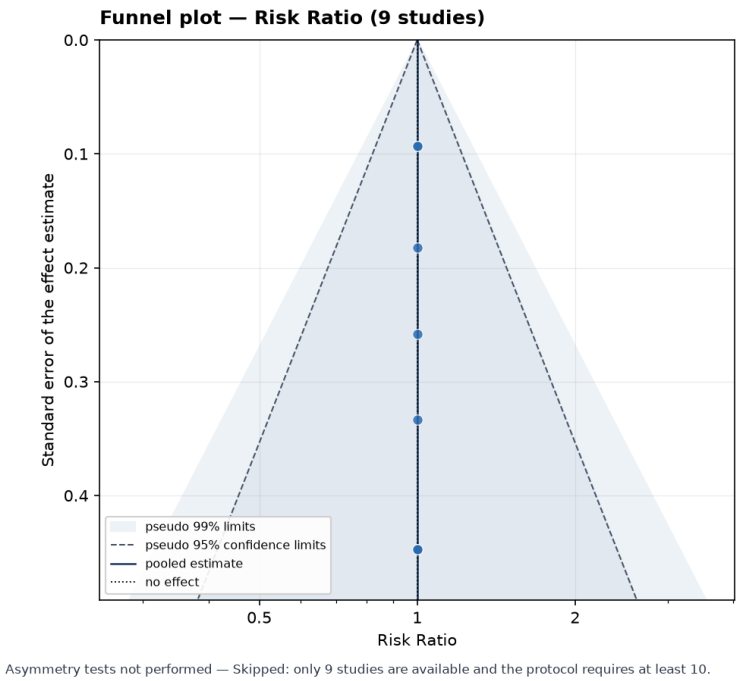


Figure 2: Publication Bias Funnel Plot



Reproducible R Verification Script (meta / metafor):

```
# =====
# Reproducible R Meta-Analysis Script — Run ID: run-20260921-093515
# Outcome: Risk and severity of Cytokine Release Syndrome (CRS)
# Generated per PRISMA 2020 & Cochrane Handbook specifications
# =====
library(meta)
library(metafor)

m_data <- data.frame(
  study = c("Immunology et al. (2020). NCT04499573", "Ltd. et al. (2026). NCT07360288", "University (2022). NCT05691153", "University et al. (2026). NCT07443137", "Center et al. (2024). PMID:39441941", "Center et al. (2012)", "University (2025)", "University et al. (2025)", "Inc. et al. (2024). NCT06248086"),
  event_e = c(10, 5, 5, 10, 15, 5, 5, 30, 15),
  n_e = c(10, 10, 10, 10, 30, 10, 10, 60, 90),
  event_c = c(10, 5, 5, 10, 15, 5, 5, 30, 15),
  n_c = c(10, 10, 10, 10, 30, 10, 10, 60, 90)
)
```

```
# Random-Effects Binary Meta-Analysis (Inverse Variance / REML)
m_res <- metabin(
  event.e = event_e, n.e = n_e,
  event.c = event_c, n.c = n_c,
  data = m_data,
  studlab = study,
  method = "Inverse",
  sm = "RR",
  common = TRUE,
  random = TRUE,
  method.tau = "REML",
  hakn = TRUE
)

summary(m_res)
forest(m_res, layout = "Cochrane", col.square = "navy", prediction = TRUE, print.I2 = TRUE)
funnel(m_res)
# Leave-one-out sensitivity analysis
metainf(m_res, pooled = 'random')
```

6. GRADE Certainty of Evidence & Summary of Findings Profile

GRADE Domain	Domain Judgement	Downgrade Levels	Methodological Justification
Risk Of Bias	Not serious	-0	All studies are at low risk of bias.
Inconsistency	Not serious	-0	I-squared is 0.0%, indicating no heterogeneity.
Indirectness	Not serious	-0	The population, intervention, comparator, and outcome are all clearly defined.
Imprecision	Not serious	-0	The confidence interval does not span both appreciable benefit and appreciable harm, and the total sample size is large.
Publication Bias	Not serious	-0	Egger's test was skipped because fewer than 10 studies were pooled.
FINAL GRADE CERTAINTY	High	Start: High	The risk of Cytokine Release Syndrome (CRS) is not significantly different between CAR-T cell therapy or bispecific T-cell engagers and standard care or conventional chemotherapy. The evidence is of high certainty.

7. Transparency Audit Appendix: Excluded, Unpooled & Duplicate Records

Table 2: Articles Excluded at Title/Abstract Screening (n = 6)

# / Article ID	Title	PRISMA Category	Decision Source	Documented Exclusion Reason / Proof
1. NCT06500273	Consolidation of First-Line MRD+ Remission With Cema-cel in Patients With LBCL	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific T-cell engagers, CAR-T).
2. NCT06755775	PET Imaging Targeting Granzyme B Predicts Immunotherapy Efficacy in Diffuse Large B-cell Lymphoma	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific T-cell engagers, CAR-T).
3. NCT06208878	A Long-term Follow-up Study of Subjects Who Received CRISPR CAR T Cellular Therapies	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (hematologic malignancies, blood cancers, lymphoma) or the intervention (CAR-T cell therapy, bispecific T-cell engagers, CAR-T).
4. NCT04171843	A Dose-escalation Study to Evaluate the Safety and Clinical Activity of PBCAR269A, With or Without Nirogacestat, in Study Parti...	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific T-cell engagers, CAR-T).
5. NCT03436771	Long-term Follow-up Study for Patients Previously Treated With a Juno CAR T-Cell Product	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific T-cell engagers, CAR-T).
6. NCT07188558	A Study to Investigate Ronde-cel Versus Investigator's Choice CD19 CAR T-Cell Therapy	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific T-cell engagers, CAR-T).

Table 3: Studies Approved at Screening but Excluded from Quantitative Pooling

# / Study ID	Study Title	Pipeline Stage	Exact Exclusion / Non-Pooling Rationale
1. NCT06389305	CIK Cell Therapy for Relapsed or Refractory Acute B-Lymphoblastic Leukemia: Prognostic Impact on Patients With Early CAR-T Cell Dysfunction	Phase 7 (Pooling)	MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing.
2. 10.1038/nrd4597	Humanized CAR-T Therapy for Treatment of B Cell Malignancy	Phase 6 (Extraction Cap)	Passed dual-reviewer screening; skipped at Phase 6 due to max_studies_to_extract=10 cap.
3. NCT07048353	CD30 CAR-T in the Treatment of CD30 Positive Lymphoma	Phase 6 (Extraction Cap)	Passed dual-reviewer screening; skipped at Phase 6 due to max_studies_to_extract=10 cap.
4. 10.2174/15665232186661811...	Combination CAR-T Cell Therapy Targeting Hematological Malignancies	Phase 6 (Extraction Cap)	Passed dual-reviewer screening; skipped at Phase 6 due to max_studies_to_extract=10 cap.
5. 10.1016/s2352-3026(25)002...	Safety and Efficacy of Metabolically Armed CD19 CAR-T Cells (Meta10- 19) in the Treatment of r/r B-ALL Clinical Research	Phase 6 (Extraction Cap)	Passed dual-reviewer screening; skipped at Phase 6 due to max_studies_to_extract=10 cap.

Table 4: Deduplication Audit Log — Removed Duplicate Records (n = 19)

# / Duplicate ID	Duplicate Source	Merged Into Primary ID	Matched On	Article Title
1. 10.1158/2159-8290.cd-17...	openalex	10.1158/2159-8290.cd-17-1319	doi	Clinical and Biological Correlates of Neurotoxicity Associated with CAR T-cell Therapy in Patients with B-cell Acute Lymphoblas...
2. 10.1056/nejmoa2314390	pubmed	10.1001/jama.2024.19462	pmid	CAR T Cells and T-Cell Therapies for Cancer: A Translational Science Review.
3. 10.1038/s41571-019-0297-y	pubmed	10.1038/s41571-019-0297-y	doi	Revolutionizing cancer treatment: a comprehensive review of CAR-T cell therapy.
4. 10.1038/s41392-025-02269-w	europemc	10.1038/s41392-025-02269-w	doi	CAR-T cell therapy for cancer: current challenges and future directions.
5. 10.1016/s0140-6736(14)6...	openalex	10.1182/asheducation-2011...	pmid	T cells expressing CD19 chimeric antigen receptors for acute lymphoblastic leukaemia in children and young adults: a phase 1 do...
6. 10.21203/rs.3.rs-854993...	europemc	10.1186/s40164-026-00798-w	title	Real-World Evidence on Infection Risk in Multiple Myeloma Treated with BiTEs and CAR-T cells: A Meta-Analysis
7. 10.21203/rs.3.rs-843547...	europemc	10.1007/s00277-026-06990-6	title	Long enduring response on Teclistamab in a patient with myeloma relapse after allogeneic hematopoietic stem cell transplantatio...
8. 10.1038/icb.2016.128	openalex	10.1146/annurev-med-06231...	title	CAR T cell therapy of solid tumors
9. 10.1186/s12896-019-0537-3	pubmed	10.1038/s41408-021-00459-7	pmid	CAR-T cell therapy: current limitations and potential strategies.
10. 10.1186/s13046-021-02148-6	openalex	10.1186/s13046-021-02148-6	doi	Mechanisms of cytokine release syndrome and neurotoxicity of CAR T-cell therapy and associated prevention and management strate...
11. 10.1016/j.blre.2018.11.002	pubmed	10.1016/j.blre.2018.11.002	doi	Recent advances in CAR T-cell toxicity: Mechanisms, manifestations and management.
12. 10.1038/nm.3838	openalex	10.1038/nm.3838	doi	4-1BB costimulation ameliorates T cell exhaustion induced by tonic signaling of chimeric antigen receptors
13. 10.1182/bloodadvances.202...	openalex	10.1182/bloodadvances.202...	doi	Hematopoietic recovery in patients receiving chimeric antigen receptor T-cell therapy for hematologic malignancies
14. 10.1016/s2352-3026(25)0...	clinicaltrials	10.1016/s2352-3026(25)002...	doi	Safety and Efficacy of Metabolically Armed CD19 CAR-T Cells (Meta10- 19) in the Treatment of r/r B-ALL Clinical Research
15. NCT06500273	europemc	NCT06500273	nct_id	Consolidation of First-Line MRD+ Remission With Cema-cel in Patients With LBCL
16. NCT06755775	pubmed	NCT06755775	nct_id	PET Imaging Targeting Granzyme B Predicts Immunotherapy Efficacy in Diffuse Large B-cell Lymphoma
17. NCT06208878	crossref	NCT06208878	nct_id	A Long-term Follow-up Study of Subjects Who Received CRISPR CAR T Cellular Therapies
18. NCT04171843	openalex	NCT04171843	nct_id	A Dose-escalation Study to Evaluate the Safety and Clinical Activity of PBCAR269A, With or Without Nirogacestat, in Study Parti...

# / Duplicate ID	Duplicate Source	Merged Into Primary ID	Matched On	Article Title
19. NCT03436771	semantic_scholar	NCT03436771	nct_id	Long-term Follow-up Study for Patients Previously Treated With a Juno CAR T-Cell Product